Establishing FSI Model Credibility for Aortic Valve Leaflet Coaptation and Stress Across Three Mechanisms: An ASME V&V 40–Guided Assessment

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Abstract

Computational models that resolve fluid–structure interaction (FSI) in the aortic root are increasingly used to evaluate transcatheter aortic valve leaflet behavior under clinical loading, yet their deployment in regulatory and design decisions hinges on transparent, mechanism-specific credibility. Following ASME V&V 40 guidance, we assessed three models for a polymeric tri-leaflet transcatheter aortic valve: Model A, a fully coupled immersed-boundary FSI solver with anisotropic hyperelastic leaflets; Model B, a partitioned arbitrary Lagrangian–Eulerian (ALE) FSI approach with nonlinear thin-shell leaflets; and Model C, a weakly coupled surrogate that prescribes leaflet kinematics to drive computational fluid dynamics (CFD). Each model was exercised against three mechanisms central to device function: M1, diastolic seal under quasi-static pressure; M2, peak systolic ejection under pulsatile flow; and M3, early-diastolic closing transient with sinus vortex–induced flutter. We adopted a private 0–7 Credibility Attainment Index (target ≥5) and scored six factors for every model × mechanism pair: numerical resolution adequacy, governing formulation, input characterization, algorithmic convergence, code governance and testing, and realism of the experimental environment used for comparison. Evidence included grid/time refinement studies; coupling tolerance and residual controls; alternative constitutive laws and coupling strategies; input calibration against biaxial test data and prescribed pressure–flow waveforms; software configuration management with regression suites and independent review; and a pulsatile bench configured to ISO 5840–concordant waveforms. For Model A, refinement altered coaptation area by 0.8% and peak stress by 2.1% in M1, with similar small changes in M2–M3; monolithic solves achieved energy imbalance ≤1.2% across mechanisms; and anisotropic material behavior was necessary to limit leaflet stress increases otherwise observed with isotropic laws (14–19%). Model B exhibited larger sensitivity to partitioning and curvature resolution (e.g., 5.4% stress change under refinement in M2) but remained within target for diastolic seal. Model C supported pressure distribution estimates during diastole (0.6% pressure map change under refinement) but diverged in ejection and closing transients where flutter and load sharing matter (stress proxies deviating by 35–38% from coupled predictions). Software test coverage ranged from 71–92% across codes, with stable regression suites. Bench waveforms reproduced physiologic targets with pressure root-mean-square deviations of 1.9 mmHg in diastole, peak flow RMSD of 18 mL/s in systole, and sinus vortex metrics within 12% of target. Scorecards (54 rows total) document per-mechanism gradations and justifications. The assessment concludes that Model A is suitable for all three mechanisms at or above the target, Model B is suitable for quasi-static seal and conditionally for ejection and closing transients with caution on thin-shell stress hot spots, and Model C is appropriate for diastolic pressure mapping only. We outline improvements aligned to the specific credibility factors to raise underperforming combinations above the target threshold.

Keywords: aortic valve FSI, V&V 40, leaflet coaptation

1. Introduction

Transcatheter aortic valves must maintain a delicate balance between hemodynamic performance and durability across the entire cardiac cycle. The success of next-generation polymeric tri-leaflet devices depends on the integrity of coaptation during diastole to suppress regurgitation, the competence of opening during peak systole to minimize gradients, and the resilience to unsteady closing loads that may precipitate fatigue damage. Computational models that resolve the mutual interaction of leaflets and blood flow can illuminate these behaviors with spatial and temporal resolution difficult to achieve experimentally. Yet their use in design selection, risk assessment, and regulatory submissions requires credible, mechanism-specific evidence following a structured framework.

ASME V&V 40 provides a risk-informed approach for assessing model credibility for medical devices by tailoring verification and validation activities to the question of interest and the specific context of use. In that spirit, this study presents a focused evaluation of three computational models applicable to a polymeric transcatheter aortic valve, with attention to tri-leaflet coaptation geometry and stress. The three models span different numerical philosophies and levels of coupling strength: a fully coupled immersed-boundary FSI solver (Model A), a partitioned ALE FSI approach with nonlinear thin shells (Model B), and a weakly coupled surrogate that prescribes leaflet motion for CFD

(Model C).

We evaluate each model against three mechanisms that jointly capture the essential leaflet mechanics under physiological conditions. Diastolic seal is treated as a quasi-static pressure loading with negligible through-flow; peak systolic ejection is examined as a pulsatile, high-Reynolds-number regime; and the early-diastolic closing transient is analyzed with emphasis on sinus-driven vortices that can excite leaflet flutter. For every model × mechanism pair, we score six credibility factors central to scientifically defensible predictions of coaptation geometry and leaflet stress. The result is a set of three per-model tables, each comprising 18 graded entries, that together summarize 54 independent credibility judgments and their documented bases.

The objective is twofold: first, to make the assumptions, numerical choices, and supporting evidence sufficiently explicit that an informed reader can judge model applicability; and second, to identify a prioritized path to elevate those model–mechanism combinations that do not meet the target credibility threshold for the intended use.

Scoping the question of interest under V&V 40 — gradation.

a.
Mechanisms not specified; global device performance treated as a single condition.

b.
Mechanisms listed but not tied to decisions; mixed outputs without clear priority.

c.
Mechanisms explicitly named with rationale; outputs aligned to each mechanism and decision.

d.
Mechanisms, outputs, and decisions fully mapped; risk-based tailoring of evidence to each mechanism.

2. Question of Interest, Context of Use, and Rationale

Question of Interest. Can computational models quantify diastolic coaptation geometry and leaflet stress in a polymeric tri-leaflet transcatheter aortic valve with sufficient credibility to inform design iteration and risk assessment for regurgitation and durability across physiologic operating conditions?

Context of Use. The intended application is comparative evaluation of leaflet mechanics among design variants within one valve platform during preclinical development. Specifically, we seek to discriminate designs that risk inadequate diastolic sealing area or exhibit elevated commissural or belly stress concentrations under clinically representative loading. The models support design ranking, identification of high-risk geometries and materials, and selection of prototypes for bench testing.

Rationale for Mechanism Decomposition. Diastolic seal under quasi-static pressure (M1) isolates the coaptation behavior in the absence of significant flow; it is pivotal to regurgitation risk and sets a baseline for stress under sustained load. Peak systolic ejection (M2) represents rapid opening, high strain-rate deformation, and transient fluid loading at elevated Reynolds number; it typifies fatigue-relevant peak stresses and dynamic load sharing. Early-diastolic closing with sinus vortex–induced flutter (M3) adds unsteady aerodynamic excitation that can perturb closure and generate stress hot-spots during impact and flutter, which are implicated in early material damage.

Decision Needs and Risk. For development-phase go/no-go decisions, the consequences of an incorrect conclusion about coaptation integrity or stress hot-spots include wasted prototyping cycles and potentially overlooking durability concerns. To align with these risks, we require that per-factor credibility reach or exceed a threshold level, here set as 5 on a 0–7 scale, for the mechanisms that drive the decision. A lower grade is acceptable for exploratory screening, but not for ranking designs with direct implications for bench testing priorities.

Decision-critical output mapping — gradation.

a.
Outputs are undefined or generic (e.g., overall performance).

b.
Outputs are enumerated but not traced to decision thresholds.

c.
Outputs have quantitative thresholds linked to design choices or risks.

d.
Outputs have quantitative thresholds and traceable acceptance criteria with sensitivity to risk.

3. Model Risk Analysis and Justification

We mapped failure modes to outputs and mechanisms to prioritize credibility investments. Insufficient diastolic coaptation area or low commissural apposition height under quasi-static load correlates with regurgitation risk; therefore, predicting coaptation geometry reliably in M1 is essential. Elevated stresses at commissures and near suture tabs, particularly during M2 and M3, relate to durability risks; hence, stress prediction across transient opening and closing is also essential. The relative risk of using a given model increases when the model omits physical effects that dominate a mechanism. For example, a weakly coupled surrogate that neglects fluid feedback on leaflet motion may suffice for quasi-static closure pressure mapping but is risky during M2 and M3, where unsteady fluid forces and flutter can dominate stresses.

We therefore examined the three models with attention to their coupling strategy, material representation, and numerical stability. Model A directly couples fluid and structural fields in a single solve, enabling robust stress prediction during rapid transients. Model B, by partitioning, introduces potential lag between fluid and structure but enables mature, high-order CFD and shell formulations at moderate cost. Model C, by prescribing kinematics, cannot capture two-way interactions and is thus less reliable where load sharing between fluid and leaflets is crucial. This risk profile influenced our scoring expectations: we anticipated that Model A would meet target thresholds across mechanisms, Model B would meet targets for M1 and be close for M2–M3, and Model C would primarily be suitable for M1.

[redacted]

Figure 1: Schematic of the three computational approaches (Model A, Model B, Model C) and the three mechanisms (M1–M3) evaluated in this study, with outputs of interest (coaptation area, commissural apposition height, residual gap, and leaflet stress) indicated alongside each mechanism.

Risk-based tailoring of evidence — gradation.

a.
No risk analysis; same evidence expected across all mechanisms.

b.
Qualitative risk analysis informs evidence depth but not scope.

c.
Quantitative risk drivers mapped to mechanisms and evidence selection.

d.
Risk-informed, mechanism-specific evidence with contingency triggers for additional testing.

4. Computational Models: Formulations and Computed Quantities

Model A: Immersed-Boundary FSI with Anisotropic Hyperelastic Leaflets. We implemented a three-dimensional, fully coupled immersed-boundary FSI solver. The blood domain is solved with an incompressible finite-volume method on a Cartesian grid with adaptive mesh refinement in the aortic root and sinus regions. The leaflets are discretized with eight-node hexahedral finite elements, and coupling is performed by spreading structural forces to the Eulerian fluid grid and interpolating fluid velocities to the Lagrangian leaflet mesh within a monolithic Newton–Krylov framework. The polymer leaflet constitutive law follows an anisotropic hyperelastic formulation calibrated to planar biaxial tests: a Fung-type strain energy density with distinct fiber directions and dispersion. What it computes. For each beat, we resolve diastolic coaptation area, commissural apposition height, and peak leaflet von Mises stress at commissures and belly during open, closing, and closed phases.

Model B: Partitioned ALE FSI with Thin-Shell Leaflets. We coupled a moving-mesh CFD solver (arbitrary Lagrangian–Eulerian finite volumes) to a nonlinear structural thin-shell solver based on a Kirchhoff–Love formulation with large rotations and geometric nonlinearity. The fluid–structure interface is matched node-to-cell-face at the leaflet midsurface; kinematic and dynamic continuity are enforced by iterative Dirichlet–Neumann coupling with Aitken under-relaxation. The membrane model includes an orthotropic hyperelastic surface law, with fiber directions aligned to the structural frame. What it computes. The model outputs transient orifice dynamics including effective orifice area during systole, coaptation line integrity at diastole, and membrane stress estimates derived from the shell midsurface stress resultants.

Model C: Weakly Coupled Surrogate (CFD with Prescribed Leaflet Kinematics). We prescribe leaflet kinematics from a separate structural simulation that neglects fluid feedback, then drive a CFD solution around these time-varying surfaces. Leaflet displacements are smoothed to prevent non-physical accelerations; the CFD resolves pressure distribution across the valve during closure and opening. Stress proxies are computed by mapping the inverse of the pressure distribution to the leaflet midsurface as loads for a static structural calculation at selected phases, acknowledging this proxy neglects transient inertial and hydrodynamic coupling effects. What it computes. We estimate closing pressure distribution, residual coaptation gap width, and stress proxies at commissures and belly.

Across all models, the geometry is a 26-mm nominal diameter tri-leaflet polymeric valve crimped and deployed in an idealized cylindrical aortic root with three sinuses. Leaflet thickness is 120 μm nominal. The heart rate is 60 bpm with a cardiac output of 5.0 L/min in the pulsatile cases; the diastolic pressure target is 80 mmHg. The fluid is represented as a Newtonian mixture with dynamic viscosity of 3.5 cP at 37°C; density is 1060 kg/m3. The systolic flow peak is approximately 400 mL/s, generating a peak Reynolds number near 4000 based on the 26-mm annulus.

Material model credibility (constitutive adequacy for leaflet tissue or polymer) — gradation.

a.
Linear elasticity or generic isotropy without calibration to material tests.

b.
Nonlinear isotropy with limited calibration to uniaxial data.

c.
Orthotropic or anisotropic nonlinear model calibrated to biaxial data with reported goodness-of-fit.

d.
Anisotropic nonlinear model with rate effects and damage, calibrated to multi-modal tests with uncertainty quantification.

5. Mechanisms of Interest and Clinical Relevance

M1: Diastolic Seal Under Quasi-Static Transvalvular Pressure. When the valve is closed, the coaptation area must be sufficient to prevent regurgitation under sustained pressure. The commissural apposition height and uniformity of contact matter because local loss of contact can seed leakage jets. The stress field under constant load informs creep and long-term deformation of polymeric materials. Because net through-flow is negligible, this mechanism isolates leaflet contact mechanics and load-bearing.

M2: Peak Systolic Ejection Under Pulsatile Flow. During rapid opening, the leaflets experience large-amplitude, high strain-rate deformations while interacting with accelerating blood flow. Peak stresses arise as the leaflets snap open and achieve large curvature near commissures. Flow transients, including vena contracta formation and jet shear layers, impose unsteady pressure distributions. Errors in load sharing prediction can bias stress hot spots and overstate or understate fatigue risk.

M3: Early-Diastolic Closing Transient with Sinus Vortex–Induced Flutter. As flow decelerates and reverses, vortices in the sinuses can excite oscillations in the leaflets prior to full contact. These unsteady excitations alter the timing and geometry of coalescence and amplify local stress concentrations, particularly if dynamic contact and pre-strain interact with hydrodynamic forcing. Flutter-like oscillations may persist for several milliseconds and leave a signature in both gap dynamics and stress peaks.

In our pulsatile bench, diastolic and systolic waveforms were tuned to ISO 5840 envelopes. The pressure trace during diastole matched the 80 mmHg target with a root-mean-square deviation of 1.9 mmHg (2.4%), while peak systolic flow reached 400 ± 18 mL/s. For closing transients, the swirl strength in each sinus was adjusted by contouring the sinus geometry and compliance elements to achieve a non-dimensional vortex formation parameter corresponding to a Strouhal number of 0.19, within 12% of a 0.21 target derived from prior in vitro visualizations.

A single sentence summarizing a factor-level finding for environmental realism, placed away from the section named for that factor: The pulsatile laboratory setup reproduced target waveforms and vortex characteristics within 2.4% in diastolic pressure RMSD, 4.5% in peak flow RMSD, and 12% in sinus swirl metrics, providing a physiologically meaningful reference against which to interpret the three mechanisms.

Representativeness of the reference environment for mechanism-specific comparisons — gradation.

a.
Nominal settings without documentation of physiological targets.

b.
Targets documented but deviations unquantified.

c.
Targets documented with quantitative deviations and uncertainty bounds.

d.
Targets documented, deviations quantified, and sensitivity of mechanism-specific metrics demonstrated.

6. Credibility Factors and Scoring Methodology (0–7 Credibility Attainment Index)

We adopted a private 0–7 Credibility Attainment Index to grade six factors for each model × mechanism pair. The six factors are: numerical resolution adequacy; governing formulation; input characterization; algorithmic convergence; code governance and testing; and realism of the experimental environment used for comparison. The target level for the intended use is 5 or greater for the relevant mechanisms. Each factor was treated independently, and no merging of factors or mechanisms was performed. Evidence was accumulated from simulation studies, software process artifacts, and bench data. The tables in the Results sections report the level assigned for each factor and mechanism, together with a concise numerical basis that is expanded upon in the prose.

Scores were computed by mapping observed practices and quantitative outcomes to the rungs described below. For algorithmic exercises (e.g., grid refinement), the mapping was based on both the breadth of the study (number of refinements, metrics tracked) and quantitative changes in outputs of interest at the finest levels. For formulation choices (e.g., material model), we emphasized the degree of physics captured and the sensitivity of outputs to plausible model alternatives. Input characterization combined calibration quality (data sources, fit errors) with uncertainty propagation for outputs. Software process health integrated coverage metrics, regression test outcomes, and independent review. Finally, representativeness of the bench setup captured the fidelity of mechanism-relevant metrics.

All scores are specific to the three mechanisms and do not imply validity outside the stated context. Where evidence was not available or was weak, the corresponding factor received a low level rather than being omitted.

Mesh and time resolution adequacy for outputs of interest — gradation.

a.
Single resolution; no systematic refinement or output tracking.

b.
One refinement in either space or time; outputs tracked qualitatively.

c.
At least two refinements in space and time; quantitative output changes below mechanism-specific tolerances.

d.
Systematic multi-parameter refinement with extrapolation to zero mesh/time step and uncertainty bounds.

7. Discretization Error: Mesh and Time-Refinement Studies by Model × Mechanism

For each model and mechanism, we conducted coordinated studies that refined both the spatial and temporal resolution by factors of approximately two. Outputs tracked included coaptation area and commissural apposition height for M1, effective orifice area and peak leaflet stress for M2, and residual coaptation gap width and peak stress for M3. We ensured that the refined solutions were feasible within available computational budgets by focusing refinement in regions of high gradients (e.g., commissures, sinuses) and during critical phases (e.g., closing onset).

Model A used a base grid of 1.2 million cells with localized refinements to 2.3 million cells; time step was reduced from 0.2 ms to 0.1 ms in the refined run. The structural mesh was refined from approximately 40k to 80k hexahedral elements per leaflet, with matching time integration. Model B employed a base ALE mesh of 1.0 million cells with 0.25 ms time step, refined to 2.0 million cells with 0.125 ms; the thin-shell mesh increased from 15k to 30k elements per leaflet. Model C's CFD mesh had a base of 0.9 million cells and was refined to 1.8 million cells; the time step decreased from 0.2 ms to 0.1 ms, while prescribed kinematics were smoothed with consistent temporal spline parameters.

The details of changes in outputs under refinement are summarized in the Results sections for each model. Briefly, for Model A the refined solution altered diastolic coaptation area by 0.8% and peak stress by 2.1% (M1), effective orifice area by 1.5% and peak stress by 3.8% (M2), and residual gap width by 1.8% with peak stress change of 2.6% (M3). For Model B, the changes were larger, with diastolic coaptation area changing by 1.4% and peak stress by 3.2% (M1), effective orifice area by 2.7% and peak stress by 5.4% (M2), and residual gap width by 2.1% with peak stress change of 6.0% (M3). For Model C, the closing pressure distribution under refinement changed by 0.6% and the residual gap proxy by 8% (M1), the pressure drop by 1.1% and stress proxy by 7.5% (M2), and vortex-induced pressure fluctuation amplitude by 4.9% with gap proxy change of 10% (M3).

These studies informed the per-mechanism scoring for numerical resolution adequacy, as reported later, without conflating them with other factors such as solver algorithmic convergence or model form. The refined changes were compared to mechanism-specific tolerances; for example, for M1, coaptation area changes below 2% and stress changes below 3% were considered acceptable for design ranking.

Temporal resolution sufficiency during rapid transients — gradation.

a.
Single time step or fixed-phase evaluations with no transient tracking.

b.
One halved time step with qualitative assessment of changes.

c.
Multiple reductions in time step with quantitative changes in transient peaks reported.

d.
Time-step extrapolation with event-aligned sampling and convergence in timing and amplitude of peaks.

8. Numerical Solver Error: Coupling Tolerances, Residual Controls, and Partitioning Stability

Algorithmic convergence was managed differently across the three models, reflecting their coupling strategies. Model A uses a monolithic Newton–Krylov framework with consistent Jacobians assembled from both fluid and structural residuals. Nonlinear iterations continue until the L2 norm of the combined residual reduces by eight orders of magnitude (1e8 relative), with line search safeguarding. Energy balance over one mechanism-specific window (e.g., one diastolic phase) is evaluated to ensure that artificial dissipation or residual imbalance does not bias outputs. Across mechanisms, Model A achieved energy imbalance not exceeding 1.2%.

Model B solves fluid and structure iteratively with Aitken relaxation and under-relaxed interface updates. The fluid subsystems target residual reductions of 1e6, whereas the structural subsystem resolves to force balance within 1e7. Interface iterations continue until normal traction and displacement mismatches fall below 1e4 in nondimensional form or a maximum of eight sub-iterations is reached. The energy drift over a cardiac phase was under 3.2% at worst (M2), with a typical range of 1.7% (M1) to 2.8% (M3).

Model C avoids two-way coupling and therefore has no interface residual to monitor. The CFD solver residual reductions were set to 1e6, and the prescribed leaflet kinematics were smoothed to avoid numerical oscillations. Because of the lack of two-way work exchange, we computed a work imbalance metric between pressure–velocity work and structural work implied by the kinematics. This metric was 4.5% for M1, 6.8% for M2, and 7.4% for M3, reflecting increasing departure from dynamic balance as transient effects grow.

Stability of partitioning in Model B during the closing transient (M3) required adaptive under-relaxation; we observed convergence within five sub-iterations for 94% of time steps in M3, with two brief intervals requiring the eight-iteration limit, which triggered a local time step reduction by 20% to restore stability. No divergence occurred. For Model A, monolithic convergence was robust, with fewer than six Newton iterations per time step in M2's most demanding phase.

A single sentence summarizing a factor-level finding for algorithmic convergence, placed away from the section named for that factor: Across mechanisms, the combined residuals and energy accounting stayed within 1.2% for the monolithic solver and within 3.2% for the partitioned solver, whereas the surrogate exhibited work imbalance approaching 7.4% when transients dominated.

Coupling tightness and interface convergence for FSI — gradation.

a.
Loose coupling without interface residual checks.

b.
Interface residuals monitored with fixed iteration counts.

c.
Adaptive coupling with residual-based stopping criteria and energy checks.

d.
Monolithic coupling or demonstrably equivalent tight partitioning with verified energy balance and robustness under transients.

9. Model Form: Governing Equations, Constitutive Laws, and Coupling Strategy

Governing Equations. All models solved incompressible Navier–Stokes for the fluid domain with no-slip at leaflet surfaces and prescribed pressure or flow boundary conditions at the ventricular and aortic ends, depending on mechanism. Structural dynamics followed large-deformation kinematics appropriate to thin polymer leaflets in either three-dimensional solid (Model A) or shell (Model B) form; Model C used a separate structural solution to generate kinematics.

Constitutive Representation. The anisotropic hyperelastic material in Model A was calibrated to planar biaxial tests of the polymer film with fiber directions aligned to manufacturing draw lines. The Fung-type model fit achieved R2 values of 0.97 in the 0–40% strain range under five loading paths; dispersion parameters were inferred from wide-angle X-ray scattering data. Model B implemented an orthotropic hyperelastic membrane law on the midsurface calibrated to the same data but lacking explicit bending stiffness representation beyond that introduced by the shell formulation. Model C used the same structural constitutive model as Model A for the kinematic prescription but, by decoupling, failed to transfer hydrodynamic effects back to the structure in real time.

Coupling Strategy and Physical Completeness. Model A's monolithic scheme captures two-way transfer of forces and displacements, including added mass and unsteady lift/drag contributions, which are essential for flutter. Model B's partitioning captures two-way coupling iteratively but introduces potential time-lag errors if under-relaxation is strong; nevertheless, it can represent unsteady pressure loading during ejection and closing with adequate sub-iterations. Model C lacks these interactions during the CFD run; the subsequent static mapping of pressure to a structural proxy neglects inertia and phase lags that are significant in M2 and M3.

Sensitivity to Model Alternatives. We performed controlled substitutions to evaluate the effect of removing anisotropy in the leaflet model. For Model A, replacing the anisotropic law with an isotropic neo-Hookean fit that matches the small-strain modulus increased peak commissural stress by 14% in M1, 19% in M2, and 17% in M3. It also reduced commissural apposition height by 0.2 mm in M1. For Model B, introducing a curvature penalty to mimic through-thickness stiffness reduced commissural stress by 6% in M2 relative to the base thin-shell, suggesting curvature resolution affects hot-spot prediction.

A single sentence summarizing a factor-level finding for governing formulation, placed away from the section named for that factor: Substituting an isotropic law in place of the calibrated anisotropic model increased peak leaflet stresses by 14–19% depending on the mechanism, demonstrating the need to retain anisotropy for design-critical stress appraisal.

Coupling strategy adequacy for mechanism physics — gradation.

a.
One-way coupling or prescribed motion for all mechanisms.

b.
Two-way coupling for selected mechanisms; others use prescribed motion.

c.
Two-way coupling for all mechanisms with documented stability controls.

d.
Monolithic or provably tight coupling for all mechanisms with evidence of capturing added mass and flutter dynamics.

10. Model Inputs: Boundary Conditions, Material Parameters, and Hemodynamic Driving Profiles

Boundary Conditions. For M1, an aortic side pressure of 80 mmHg was prescribed with a ramp to eliminate transients; the ventricular side was set to zero gauge pressure. For M2 and M3, measured bench waveforms were imposed as inlet or outlet boundary conditions as appropriate: a ventricular inflow waveform for systolic ejection peaking at 400 mL/s with a Womersley parameter of 16 based on annulus radius and cycle frequency; and a ventricular side pressure decay for early diastole that generated reverse flow and sinus vortices consistent with visualized patterns in the bench. Sensitivity to ±5% changes in pressure amplitude and timing was examined for all models.

Material Parameters. Leaflet materials were calibrated from planar biaxial tests of the polymer film at 37°C across five loading paths (equi-biaxial, fiber-dominant, cross-fiber dominant, and two off-axis cases). The anisotropic hyperelastic parameters in Model A were fit using nonlinear least squares with parameter bounds from prior draw ratio characterization; 95% confidence intervals for principal parameters were under 8%. Model B's orthotropic membrane parameters were fit to match areal strain energy surfaces from the same data set; bending stiffness was tuned to match three-point out-of-plane deflection tests on strips. Thickness was measured at 120 ± 5 µm across 24 locations on each leaflet.

Hemodynamic Driving Profiles. The bench-provided waveforms were low-pass filtered at 50 Hz to remove sensor noise. We provided the same waveforms to the CFD boundaries in Models A–C to ensure comparability. For Model C, prescribed leaflet kinematics were obtained from a structural simulation driven by these waveforms, then smoothed with cubic splines to remove jitter; we verified that the smoothing did not shift closure timing by more than 1.5 ms.

Uncertainty Propagation. We propagated uncertainties in material parameters (95% CI) and boundary waveform amplitudes (±2 mmHg in diastole, ±18 mL/s peak in systole) to outputs using linearized sensitivity analysis for Models A and B and Monte Carlo sampling (N=100) for Model C's static stress proxies. The variations in outputs are reported in the Results sections. These propagate to uncertainty bounds on coaptation area, commissural apposition height, residual gap, and peak stresses.

A single sentence summarizing a factor-level finding for inputs, placed away from the section named for that factor: Across mechanisms, uncertainties in calibrated material parameters and measured pressure–flow waveforms propagated to variations of ±3.4% to ±8.2% in coaptation or stress outputs depending on the model and mechanism, with the largest spread observed for the thin-shell stress under closing transients.

$$\text{CI\_\{score\}} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i} \tag{1}$$

$$\text{\textbackslash varepsilon\_\{rel\}} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i} \tag{2}$$

$$\text{U\_\{val\}} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i} \tag{3}$$

Input characterization and propagation to outputs — gradation.

a.
Nominal inputs with no calibration or propagation.

b.
Calibrated inputs from limited data; no uncertainty propagation.

c.
Calibrated inputs with uncertainty bounds; sensitivity analysis to key outputs.

d.
Full uncertainty propagation to outputs with sensitivity ranking and mechanism-specific tolerances.

11. Software Quality Assurance: Configuration Management, Regression Tests, and Independent Review

Configuration Management and Traceability. All three models were developed under Git-based version control with tagged releases corresponding to each set of mechanism runs. Commit messages referenced issue tracker tickets and documented parameter files for repeatability. Continuous integration pipelines built and tested the code on Linux environments with standardized compilers.

Regression and Unit Testing. Model A's code base included 212 regression tests covering constitutive updates, immersed-boundary coupling kernels, and representative FSI scenarios; line coverage was 86% and branch coverage 94% measured by gcov. Static analysis (clang-tidy) flagged three medium-severity issues that were resolved before running mechanism studies. Model B's code base reached 71% line and 83% branch coverage across 157 regression tests; two low-severity issues remained open (documentation gaps on a boundary update routine and a missing assertion in a utility), both unrelated to core solvers. Model C integrated a commercial CFD package; unit tests for custom kinematics wrappers and data mapping achieved 92% line coverage across 64 tests; the commercial solver's internal coverage is not disclosed, but vendor release notes and validation summaries were archived.

Independent Review. For Model A, an external expert from a separate institution reviewed the immersed-boundary coupling and material calibration approach; two recommendations (documenting Jacobian assembly and expanding biaxial paths) were implemented. Model B underwent internal peer review focusing on partitioning convergence controls and thin-shell bending representation. Model C's wrapper code was reviewed internally; due to limited access to the commercial solver internals, review focused on the kinematics interface and mapping routines.

Issue Resolution and Reproducibility. All mechanism runs were scripted with fixed random seeds (for any stochastic sampling in Model C) and logged on a compute cluster with environment modules recorded. Docker images were created for Model A and Model B to facilitate re-running; Model C's commercial solver required a licensed host and was run in a controlled VM environment.

A single sentence summarizing a factor-level finding for code governance and testing, placed away from the section named for that factor: Project repositories achieved 71–92% test coverage with stable regression suites and documented external review for the monolithic solver, providing transparency and repeatability across the three models.

Software process maturity and independent review — gradation.

a.
Ad hoc scripts and local builds; no tests or reviews.

b.
Basic version control and some unit tests; no independent review.

c.
Documented CI, regression tests with coverage metrics, and internal review.

d.
Mature CI/CD, high coverage, static analysis, and independent external review with documented resolution.

12. Test Conditions: Pulsatile Bench Setup, Pressure–Flow Waveforms, and Environmental Controls

Pulsatile Bench Setup. A commercial pulse duplicator was configured with a silicone aortic root featuring three sinuses matching the 26-mm valve annulus. Compliance and resistance elements were tuned to achieve physiologic left ventricular pressure–volume loops. The working fluid was a water–glycerin mixture adjusted to 3.5 ± 0.1 cP at 37.0 ± 0.2°C, monitored by inline sensors. Pressure transducers were calibrated before each run (0.5% full-scale accuracy), and electromagnetic flowmeters captured ventricular inflow and aortic outflow waveforms at 1 kHz sampling frequency.

Waveform Matching and Uncertainty. Target waveforms for M1, M2, and M3 were established from ISO 5840 envelopes and prior in vitro measurements. After tuning, the diastolic pressure waveform (M1) matched the 80 mmHg target with RMSD of 1.9 mmHg (2.4%); the systolic peak flow (M2) reached 400 ± 18 mL/s with RMSD of 18 mL/s (4.5%); and sinus vortex metrics (M3), computed from planar particle image velocimetry at 500 Hz, matched target swirl strength and vortex formation number within 12%. Environmental drift during runs was minimal: temperature stayed within ±0.2°C and viscosity within ±0.1 cP.

Boundary Condition Extraction for Models. Pressure and flow waveforms were low-pass filtered (50 Hz) and phase-aligned before use as boundary conditions in the models. For M3, reverse flow onset timing was aligned within 2 ms between bench and simulations. Leaflet kinematics were imaged with high-speed video (1000 fps) and used qualitatively to confirm closure timing; however, these images were not used to directly impose motion except in the separate structural simulation that drives Model C.

Equipment Repeatability and Samples. Three repeated runs per setting were conducted to estimate waveform repeatability. Physical prototypes used in the bench were preconditioned for 50 cycles before data acquisition.

Mention of test samples in passing without concluding a finding: The availability of three physical prototype valves for the bench runs constrained repeated testing across mechanisms and limited destructive post-test characterization of the same specimens.

Environmental control and waveform fidelity — gradation.

a.
Uncontrolled environment; waveform targets unmet.

b.
Basic control of temperature/viscosity; partial waveform fidelity.

c.
Tight control of environment with quantitative waveform fidelity and uncertainty.

d.
Tight control plus mechanism-specific flow features (e.g., sinus vortices) matched within quantified tolerances.

13. Test Samples

Three production-intent polymeric tri-leaflet valves of 26-mm size were used on the pulsatile bench. Devices originated from a single lot and underwent standard preconditioning before data collection. Specimens were mounted in a standardized test fixture with axial alignment confirmed via fluoroscopy. Post-test inspection consisted of visual examination and dimensional checks; destructive sectioning was deferred to preserve the specimens for subsequent endurance testing.

14. Results: Factor-by-Factor Scorecards for Model A

We summarize quantitative evidence for Model A across mechanisms before presenting the score table. For M1 (diastolic seal), refining the fluid grid from 1.2 to 2.3 million cells and halving the time step from 0.2 to 0.1 ms changed coaptation area by 0.8% and peak leaflet von Mises stress by 2.1%; commissural apposition height changed by 0.1 mm. Monolithic nonlinear solves reduced the combined residual to 1e8 and achieved energy imbalance of 0.6%. Substituting an isotropic neo-Hookean law for the calibrated anisotropic model increased peak commissural stress by 14% and reduced commissural apposition height by 0.2 mm. Uncertainty propagation from material parameter confidence intervals and ±2 mmHg diastolic pressure variation resulted in ±3.4% variation in coaptation area and ±5.6% in peak stress. Code process metrics included 86% line and 94% branch coverage across 212 regression tests, with zero open severity-1 defects. The bench diastolic pressure trace matched the 80 mmHg target with RMSD of 1.9 mmHg (2.4%) at 37.0 ± 0.2°C and viscosity 3.5 ± 0.1 cP.

For M2 (peak systolic ejection), refinement changed effective orifice area by 1.5% and peak leaflet stress by 3.8%. The monolithic solver maintained energy imbalance at 0.9% with combined residuals under 1e8. Using the isotropic alternative increased peak stress by 19% and reduced effective orifice area by 4%. Input uncertainty from material parameters and ±18 mL/s variation in peak flow yielded ±4.2% variation in peak stress. Software and bench metrics were as above; systolic peak flow matched 400 mL/s with RMSD of 18 mL/s (4.5%).

For M3 (early-diastolic closing transient), refinement changed residual coaptation gap width by 1.8% and peak stress by 2.6%. The monolithic solver achieved energy imbalance of 1.2% under transient closing loads. The isotropic substitution increased peak stress by 17% and delayed closure by 3 ms. Input uncertainties, including waveform timing variability within 2 ms, produced ±6.1% variation in peak stress. Bench sinus vortex metrics matched target within 12%; Strouhal number was 0.19 versus a 0.21 target.

Interpreting 0–7 index levels for mechanism-specific credibility — gradation.

a.
Levels 0–1: insufficient evidence; exploratory only.

b.
Levels 2–3: partial evidence with notable gaps; not decision grade.

c.
Levels 4–5: mechanism-aligned evidence with quantified limits; conditionally decision grade.

d.
Levels 6–7: comprehensive, mechanism-tailored evidence with quantified uncertainty; decision grade.

15. Results: Factor-by-Factor Scorecards for Model B

For Model B (partitioned ALE with thin-shell leaflets), mechanism-specific evidence is as follows. In M1, refinement from 1.0 to 2.0 million cells and halving the time step from 0.25 to 0.125 ms changed coaptation area by 1.4% and peak stress by 3.2%; commissural apposition height changed by 0.2 mm. Interface iterations achieved traction–displacement mismatch below 1e4 in under five sub-iterations for 96% of time steps; energy drift over diastole was 1.7%. Sensitivity to isotropy was similar to Model A but moderated by the shell formulation; switching to isotropic increased commissural stress by 12% in M1. Input uncertainty propagation produced ±4.7% variation in coaptation area and ±6.5% in peak stress under ±2 mmHg pressure variation. Software process metrics included 71% line and 83% branch coverage across 157 regression tests, with two open low-severity issues. Bench diastolic conditions were as described previously.

In M2, refinement changed effective orifice area by 2.7% and peak leaflet stress by 5.4%. The partitioned scheme maintained energy imbalance at 3.2% worst-case with residual targets of 1e6 (fluid) and 1e7 (structure); adaptive under-relaxation enabled convergence within five sub-iterations for 93% of time steps. Introducing a curvature penalty reduced commissural stress by 6%, indicating sensitivity to bending representation; isotropic substitution increased stress by 16%. Input uncertainty (±18 mL/s peak flow, material CI) induced ±6.9% variation in peak stress. Bench systolic waveform RMSD was 18 mL/s (4.5%).

In M3, refinement changed residual gap width by 2.1% and peak stress by 6.0%. Energy drift during closing transients was 2.8%. Isotropic substitution increased peak stress by 15% and closure timing shifted by 4 ms. Uncertainty propagation in closing parameters (waveform timing ±2 ms) and material CI produced ±8.2% variation in peak stress. Sinus vortex metrics were within 12% of target.

These outcomes reflect the trade-offs inherent in partitioning and shell formulations. While M1 meets target levels comfortably, M2 and M3 approach the threshold in several factors and highlight areas for improvement, including enhanced curvature representation and tighter coupling during rapid transients.

Thin-shell adequacy for leaflet bending and curvature hot spots — gradation.

a.
Membrane-only shell with no validation of curvature effects.

b.
Shell with ad hoc bending stiffness; limited checks on curvature.

c.
Nonlinear shell with bending tuned against out-of-plane tests; sensitivity to curvature reported.

d.
Nonlinear shell with validated bending against multiple tests; curvature-driven stress hot spots benchmarked.

16. Results: Factor-by-Factor Scorecards for Model C

For Model C (weakly coupled surrogate), the evidence aligns with the expectation that diastolic pressure mapping is its strongest application. In M1, refining the CFD mesh from 0.9 to 1.8 million cells and halving the time step from 0.2 to 0.1 ms changed the closing pressure distribution by 0.6% and the residual gap proxy by 8%. Solver residuals reduced to 1e6 in CFD runs, and work imbalance between pressure–velocity work and structural work implied by prescribed kinematics was 4.5%. Input uncertainties (±2 mmHg, material CI in the structural pre-solve) yielded ±3.8% variation in the residual gap proxy and ±5.2% in stress proxies. Test process metrics for the wrappers achieved 92% line coverage across 64 tests; commercial solver internals were not accessible. Bench diastolic conditions match those reported earlier.

In M2, refinement changed the pressure drop by 1.1% and stress proxies by 7.5%. Work imbalance increased to 6.8%. Stress proxies deviated by 35% from Model A's peak stress predictions when using identical boundary waveforms and geometry, reflecting the absence of two-way coupling; effective orifice area tracking was qualitative only. Input uncertainty (±18 mL/s) induced ±7.9% variation in stress proxies. Bench systolic waveform fidelity was as above.

In M3, vortex-induced pressure fluctuation amplitude changed by 4.9% with refinement, and gap proxies changed by 10%. Work imbalance reached 7.4%. Stress proxies deviated by 38% compared to Model A due to neglected flutter and impact dynamics. Input uncertainty from ±2 ms timing and material CI induced ±10.5% variation in stress proxies. Bench sinus vortex metrics were within 12% of target but could not be faithfully imposed through kinematic prescription when feedback was absent.

These observations motivated high scores for M1 in numerical resolution and environment realism factors, moderate scores for SQA and inputs, and low scores in formulation and algorithmic convergence for M2–M3, consistent with the model's design scope.

Appropriateness of surrogate approaches for mechanism-limited questions — gradation.

a.
Surrogate used indiscriminately across mechanisms.

b.
Surrogate used with awareness of limits; minimal quantification.

c.
Surrogate limited to mechanisms where omitted physics are negligible; deviations quantified.

d.
Surrogate tightly benchmarked against coupled models for the intended mechanism with uncertainty bounds.

17. Discussion: Alignment with Context of Use and Decision-Making

The credibility assessment demonstrates that Model A, a fully coupled immersed-boundary FSI solver with anisotropic hyperelastic leaflets, consistently meets or exceeds the target threshold across all three mechanisms. For M1, the small changes under refinement (0.8% in coaptation area; 2.1% in stress) and negligible energy imbalance (0.6%) warrant confidence in diastolic coaptation predictions and stress localization. For M2 and M3, while transient demands are higher, the solver maintained energy imbalance within 1.2% and delivered sensitivity evidence that retaining anisotropy is essential (14–19% stress increases avoided). These attributes support the use of Model A for design ranking with regard to coaptation integrity and stress hot spots.

Model B, by contrast, presents a mixed picture. It achieves acceptable credibility in M1 and comes close for M2 and

M3. The principal limitations arise from thin-shell curvature representation and the lag introduced by partitioning during rapid transients. The thin-shell formulation without through-thickness stress resolution appears to over-simplify commissural curvature, as evidenced by a 6% change in commissural stress upon introducing a curvature penalty. Nonetheless, discretization changes remain moderate (e.g., 5.4% stress in M2), and energy drift is controlled (≤3.2%). These features suggest that Model B can support comparative evaluation in M1 and serve as a screening tool in M2–M3 if interpreted with caution, particularly for localized stress estimates.

Model C shows credible behavior for M1 but is not suitable for decision making in M2 or M3. The stress proxies' divergence from the fully coupled predictions (35–38%) during fast opening and closing undermines confidence in fatigue-relevant stress mapping. The work imbalance metric, which rises from 4.5% (M1) to 7.4% (M3), further highlights the lack of dynamic consistency. However, for M1 diastolic pressure distribution and gap estimation, the surrogate's small refinement-induced changes (0.6%) and manageable input-driven variability (±3.8–5.2%) can be acceptable when used to flag designs at risk of leakage before committing to higher-fidelity FSI runs.

The bench setup offers mechanism-specific realism: diastolic pressure RMSD of 1.9 mmHg (2.4%), systolic flow RMSD of 18 mL/s (4.5%), and sinus vortex characteristics within 12% of targets. These quantitative metrics lend credence to boundary waveforms used by all models, while clarifying that transient closing features (M3) can be particularly sensitive to swirl intensity and phase. Where the models differ most—flutter and dynamic contact—Model A better maps onto the mechanism, whereas Model C lacks key physics. These distinctions match the intended context of use statements and support mechanism-specific selection of modeling tools.

A single sentence summarizing a factor-level finding for numerical resolution adequacy, placed away from its titled section: For the outputs that drive decisions, refining space and time shifted Model A's predictions by less than 4% across mechanisms, while Models B and C remained below about 6% and 10%, respectively, on their most sensitive quantities.

A single sentence summarizing a factor-level finding for input characterization, placed away from its titled section: Variations in material parameters and waveform amplitudes within measured bounds translated to output spreads of roughly ±3–6% for Model A, ±5–8% for Model B, and up to ±10% for Model C's proxies, underscoring the distinct robustness of each approach.

A single sentence summarizing a factor-level finding for governing formulation, placed away from its titled section: Retaining anisotropy in the leaflet model consistently reduced predicted extreme stresses by more than ten percentage points relative to isotropic alternatives, which is material to design selection decisions.

A single sentence summarizing a factor-level finding for algorithmic convergence, placed away from its titled section: The monolithic approach kept energy and residual balances tightly controlled (≤1.2%), whereas partitioned coupling remained within a few percent and the surrogate's imbalance grew with transient severity, aligning with expectations for each mechanism.

A single sentence summarizing a factor-level finding for code governance and testing, placed away from its titled section: Test coverage between seventy and over ninety percent and stable regression suites across repositories enabled reproducible reruns and transparent traceability, which supported consistent scoring across mechanisms.

A single sentence summarizing a factor-level finding for environmental realism, placed away from its titled section and without using the canonical factor name: The laboratory environment produced stable physiologic conditions and waveforms close to target, allowing meaningful calibration and comparison without masking mechanism-specific differences.

Decision mapping from score levels to model selection — gradation.

a. No mapping; scores not used for decisions.

b. Informal mapping; ad hoc thresholds.

c. Documented thresholds tied to mechanisms; conditional acceptance rules.

d. Formal decision matrix linking scores to context-of-use selection and contingency testing.

18. Limitations and Planned Improvements Focused on Credibility Factors

A key limitation in the present exercise is the absence of three-dimensional imaging of leaflet coaptation lines to directly compare with predicted coaptation area and apposition height. While pressure and flow waveforms were tightly matched, the mechanistic validation of local gap geometry during M3 would benefit from high-speed stereo digital image correlation with transparent root models. The availability of only three physical prototypes limited destructive post hoc material testing on the same specimens to reconcile any evolution of properties with initial calibrations.

On numerical resolution, Model B's thin-shell discretization around commissures can be refined and supplemented with higher-order elements to better capture curvature without incurring excessive cost. An adaptively refined shell mesh driven by curvature indicators is under development. For solver algorithms, we plan to integrate a quasi-Newton interface acceleration scheme in Model B to tighten coupling during M2–M3 and reduce energy drift toward the sub-2% range. Model C will incorporate a reduced-order fluid–structure feedback loop during M3 to mitigate work imbalance and improve stress proxy fidelity, though full parity with coupled FSI is not anticipated.

Regarding formulation, we will extend the anisotropic material model in Model A to include rate effects and damage accumulation parameters calibrated to cyclic tests, advancing toward fatigue-relevant predictions. Model B will incorporate a through-thickness shear correction for regions of high curvature and an explicit contact model during coaptation to improve M1–M3 stress distributions. For input characterization, we will expand uncertainty propagation to include leaflet thickness variability measured via micro-CT on full devices and heartbeat-to-heartbeat waveform variability observed in the bench.

In software process, we aim to raise Model B's test coverage above 80% and add an external review focused on the ALE mesh motion and shell bending routines. For Model C, wrapper tests will be extended to include end-to-end checks of the kinematics mapping and pressure-proxy structural solution. On the bench side, we will enhance vortex quantification in M3 using volumetric three-component velocimetry to improve the match to mechanism-specific swirl metrics.

Collectively, these planned improvements target the specific factors and mechanisms where scores fell below or near the target, with the goal of raising M2–M3 for Model B and expanding Model C's reliable use in M1 without overinterpreting its proxies elsewhere.

Continuous improvement loop aligned to credibility gaps — gradation.

a.
No plan to address low-scoring factors.

b.
General plans not tied to mechanisms or factors.

c.
Specific improvements mapped to low-scoring factors and mechanisms.

d.
Funded roadmap with milestones, re-scoring triggers, and prospective evidence generation.

19. Conclusions

We conducted a mechanism-specific credibility assessment of three computational approaches for a polymeric tri-leaflet transcatheter aortic valve under ASME V&V 40 guidance. The fully coupled immersed-boundary FSI model with anisotropic hyperelastic leaflets (Model A) attained the target credibility level across diastolic seal, peak systolic ejection, and early-diastolic closing with flutter. The partitioned ALE FSI with thin shells (Model B) met the target for diastolic seal and approached it for the two transients, identifying targeted areas for improvement in curvature representation and coupling tightness. The weakly coupled surrogate (Model C) achieved adequate credibility for diastolic pressure mapping but did not meet the threshold for peak ejection or closing transients where two-way coupling is essential.

The assessment relies on quantified evidence for numerical resolution adequacy, solver convergence, formulation sensitivity, input characterization, software process rigor, and bench waveform realism. Three per-model tables enumerate 54 graded combinations, ensuring that low-evidence areas are scored rather than omitted. The approach demonstrates how mechanism decomposition and factor-specific evidence can guide model selection for design decisions. Future work will expand material models to rate-dependent behavior, enhance partitioned coupling stability, and augment bench measurements of closing vortices to further strengthen mechanism alignment.

A single sentence summarizing a factor-level finding for software process, placed away from its titled section: Across the codes, we maintained disciplined configuration control, achieved high automated test coverage, and documented independent review where feasible, enabling repeatable evidence generation for all mechanisms.

Closure and traceability of credibility claims — gradation.

a.
Claims are asserted without traceable evidence.

b.
Claims refer to evidence but lack explicit linkage.

c.
Claims linked to specific quantitative evidence and score rows.

d.
Claims fully traceable to mechanism-specific evidence with archived artifacts and rerun instructions.

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Table 1: Credibility attainment index for Model A (Immersed-Boundary FSI with Anisotropic Hyperelastic Leaflets) across three mechanisms.

Credibility factor	Level	Basis
Discretization error - Model A - M1: Diastolic Seal	6	Refinement (1.2M2.3M cells; 0.20.1 ms) changed coaptation area by 0.8% and peak stress by 2.1%.
Numerical solver error - Model A - M1: Diastolic Seal	6	Monolithic residuals to 1e8; energy imbalance over diastole 0.6%.
Model form - Model A - M1: Diastolic Seal	6	Replacing anisotropy with isotropy raised peak commissural stress by 14% and reduced apposition height by 0.2 mm.
Model inputs - Model A - M1: Diastolic Seal	6	Uncertainty propagation yielded ±3.4% in coaptation area and ±5.6% in peak stress under ±2 mmHg pressure.
Software quality assurance - Model A - M1: Diastolic Seal	6	212 regression tests; 86% line and 94% branch coverage; no Sev-1 defects.
Test conditions - Model A - M1: Diastolic Seal	6	Bench diastolic 80 mmHg with RMSD 1.9 mmHg (2.4%); 37.0 ± 0.2°C; 3.5 ± 0.1 cP.
Discretization error - Model A - M2: Peak Systolic Ejection	5	Refinement changed effective orifice area by 1.5% and peak leaflet stress by 3.8%.
Numerical solver error - Model A - M2: Peak Systolic Ejection	5	Combined residuals <1e8; energy imbalance 0.9% during rapid opening.
Model form - Model A - M2: Peak Systolic Ejection	6	Isotropic substitute increased peak stress by 19% and reduced effective orifice area by 4%.
Model inputs - Model A - M2: Peak Systolic Ejection	5	±18 mL/s flow and material CI induced ±4.2% variation in peak stress.
Software quality assurance - Model A - M2: Peak Systolic Ejection	6	Same code base: 86%/94% coverage; 212 tests; external review completed.
Test conditions - Model A - M2: Peak Systolic Ejection	5	Peak flow 400 ± 18 mL/s; RMSD 18 mL/s (4.5%) against target waveform.
Discretization error - Model A - M3: Early-Diastolic Closing	5	Refinement changed residual gap width by 1.8% and peak stress by 2.6%.
Numerical solver error - Model A - M3: Early-Diastolic Closing	5	Energy imbalance 1.2% under closing transients; residuals to 1e8.
Model form - Model A - M3: Early-Diastolic Closing	5	Isotropic law increased peak stress by 17% and delayed closure by 3 ms.
Model inputs - Model A - M3: Early-Diastolic Closing	5	Waveform timing ±2 ms and material CI gave ±6.1% variation in peak stress.
Software quality assurance - Model A - M3: Early-Diastolic Closing	6	Mature CI; 212 tests; 86%/94% coverage; no critical open issues.
Test conditions - Model A - M3: Early-Diastolic	5	Sinus swirl metrics within 12% of target; Strouhal 0.19 vs 0.21.

Table 2: Credibility attainment index for Model B (Partitioned ALE FSI with Thin-Shell Leaflets) across three mechanisms.

Credibility factor	Level	Basis
Discretization error - Model B - M1: Diastolic Seal	5	Refinement (1.0M2.0M; 0.250.125 ms) changed coaptation area by 1.4% and peak stress by 3.2%.
Numerical solver error - Model B - M1: Diastolic Seal	5	Interface mismatch <1e4; energy drift 1.7% over diastole.
Model form - Model B - M1: Diastolic Seal	5	Isotropic substitution raised commissural stress by 12%; thin shell captured apposition height within 0.2 mm.
Model inputs - Model B - M1: Diastolic Seal	5	±2 mmHg and material CI yielded ±4.7% coaptation area and ±6.5% peak stress variation.
Software quality assurance - Model B - M1: Diastolic Seal	5	157 regression tests; 71% line and 83% branch coverage; two low-severity issues open.
Test conditions - Model B - M1: Diastolic Seal	6	Diastolic 80 mmHg with RMSD 1.9 mmHg (2.4%); environment 37.0 ± 0.2°C; 3.5 ± 0.1 cP.
Discretization error - Model B - M2: Peak Systolic Ejection	4	Refinement changed effective orifice area by 2.7% and peak stress by 5.4%.
Numerical solver error - Model B - M2: Peak Systolic Ejection	4	Residual targets met; energy imbalance worst-case 3.2% during opening.
Model form - Model B - M2: Peak Systolic Ejection	4	Isotropic raise in peak stress 16%; curvature penalty lowered commissural stress by 6%.
Model inputs - Model B - M2: Peak Systolic Ejection	4	±18 mL/s and material CI produced ±6.9% variation in peak stress.
Software quality assurance - Model B - M2: Peak Systolic Ejection	5	Same code base: 71%/83% coverage; 157 tests; stable CI.
Test conditions - Model B - M2: Peak Systolic Ejection	5	Peak flow 400 ± 18 mL/s; RMSD 18 mL/s (4.5%).
Discretization error - Model B - M3: Early-Diastolic Closing	4	Refinement changed residual gap width by 2.1% and peak stress by 6.0%.
Numerical solver error - Model B - M3: Early-Diastolic Closing	4	Energy drift 2.8%; 94% of steps converged within five sub-iterations.
Model form - Model B - M3: Early-Diastolic Closing	4	Isotropy increased peak stress by 15% and delayed closure by 4 ms.
Model inputs - Model B - M3: Early-Diastolic Closing	4	Waveform timing ±2 ms and material CI yielded ±8.2% variation in peak stress.
Software quality assurance - Model B - M3: Early-Diastolic Closing	5	CI in place; 157 tests; 71%/83% coverage; no critical defects.
Test conditions - Model B - M3: Early-Diastolic Closing	5	Sinus vortex metrics within 12% of target; Strouhal 0.19 vs 0.21.

Table 3: Credibility attainment index for Model C (Weakly Coupled Surrogate: CFD with Prescribed Leaflet Kinematics) across three mechanisms.

Credibility factor	Level	Basis
Discretization error - Model C - M1: Diastolic Seal	5	Refinement (0.9M1.8M; 0.20.1 ms) changed closing pressure distribution by 0.6% and gap proxy by 8%.
Numerical solver error - Model C - M1: Diastolic Seal	4	CFD residuals to 1e6; work imbalance 4.5% under quasi-static closure.
Model form - Model C - M1: Diastolic Seal	3	Prescribed kinematics omit two-way feedback; stress proxy deviates 22% from coupled prediction.
Model inputs - Model C - M1: Diastolic Seal	5	±2 mmHg and material CI gave ±3.8% gap proxy and ±5.2% stress proxy variation.
Software quality assurance - Model C - M1: Diastolic Seal	5	92% line coverage for wrappers; 64 tests; commercial solver validation archived.
Test conditions - Model C - M1: Diastolic Seal	5	Bench diastolic 80 mmHg with RMSD 1.9 mmHg (2.4%); stable environment.
Discretization error - Model C - M2: Peak Systolic Ejection	4	Refinement changed pressure drop by 1.1% and stress proxies by 7.5%.
Numerical solver error - Model C - M2: Peak Systolic Ejection	3	Work imbalance 6.8% under rapid opening; no interface residuals available.
Model form - Model C - M2: Peak Systolic Ejection	2	Stress proxies deviated 35% from coupled predictions; no added-mass effects captured.
Model inputs - Model C - M2: Peak Systolic Ejection	4	±18 mL/s and material CI gave ±7.9% variation in stress proxies.
Software quality assurance - Model C - M2: Peak Systolic Ejection	5	Wrapper tests 92% coverage; stable CI; vendor notes archived.
Test conditions - Model C - M2: Peak Systolic Ejection	4	Peak flow 400 ± 18 mL/s; mapping to surrogate retained timing but limited dynamic fidelity.
Discretization error - Model C - M3: Early-Diastolic Closing	3	Refinement changed vortex pressure fluctuation amplitude by 4.9% and gap proxies by 10%.
Numerical solver error - Model C - M3: Early-Diastolic Closing	3	Work imbalance 7.4% under closing transients; no coupling stability metrics.
Model form - Model C - M3: Early-Diastolic Closing	2	Stress proxies deviated 38% from coupled model; flutter and impact neglected.
Model inputs - Model C - M3: Early-Diastolic Closing	3	±2 ms timing and material CI produced ±10.5% variation in stress proxies.
Software quality assurance - Model C - M3: Early-Diastolic Closing	5	Wrapper CI and tests stable (92% coverage); commercial core not inspectable.
Test conditions - Model C - M3: Early-Diastolic Closing	4	Sinus vortex metrics within 12% in bench; surrogate could not fully impose feedback-driven features.